The Matplotlib User's Guide

John Hunter

December 11, 2004

2.10 Spectral analysis 36
2.11 Axes properties 38
2.12 Legends and tables 38
2.13 Navigation 38
2.13.1 Classic toolbar 38
2.13.2 toolbar2 39
2.14 Event handling 39

Chapter 1

Introduction

```
dt = 0.01
t = arange(0,10,dt)
nse = randn(len(t))
r = exp(-t/0.05)

cnse = conv(nse, r) dt
cnse = cnse
```

```
dt = 0.01;
t = [0:dt:10];
nse = randn(size(t));
r = exp(-t/0.05);

cnse = conv(nse, r) dt;
```

IDE	GUI	Backends and Options
idle	Tkinter	Works best with TkAgg if idle is launched with -n
pycrust	WX	Works best with WX/WXAgg
Scintilla and SciTE	GTK	

```
/home/jdhunter/python/projects/matplotlib/examples
>>> run simple_plot.py
>>> title('a new title', color='r')
```

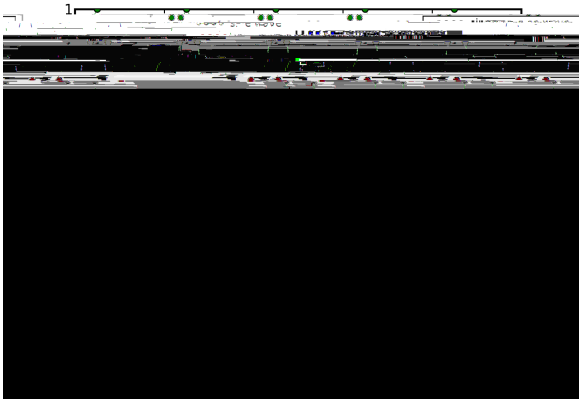


Figure 2.4:

```
>>> lines = plot(t, s1)
>>> set(lines, markersize=15, marker='d', \
...     markerfacecolor='g', markeredgecolor='r')
```

```
title(
```

2.4.2 Loading and saving binary data

ASCII is bloated and slow for working with large arrays, and so binary data should be used if performance

or use set for the same purpose

```
>>> set(fig , facecolor=fig9(e50233.26Tf.e)-5e)-23(r)-3211.9(e5.00Bfme)-bpese
```

```
ylabel('Undamped')
```

Likewise, to create two columns and one row of axes, you would use `subplot(121)` to create the left and `subplot(122)` to create the right axes. If the total number of axes exceeds single digits, use co separated arguments to subplot. For example, the lower right panel of a 3 x 4 grid of axes is created `subplot(3,4,12)`. matplotlib uses matlab style indexing in creating figures and axes, so `subplot(3,` is the first subplot, not `subplot(3,4,0)`.

2.5.3 axes

```
# instance
```

- `mlab.window_none` - no windowing
- `mlab.window_hanning`



Figure 2.15:

```
family = ['serif', 'sans-serif', 'cursive', 'fantasy',
```

Chapter 4

Collections

Chapter 5

Tick locators and formatters

The `matplotlib.ticker`

```
# but we'll override the default with our custom locators and
# formatters
ax.xaxis.set_major_
```

Listing 6.1: Fill the area between two curves; see Figure 6.1

```
from pylab import
x1 = arange(0, 2, 0.01)
y1 = sin(2 pi x1)
y2 = sin(4 pi x1) + 2

# reverse x and y2 so the polygon fills
```

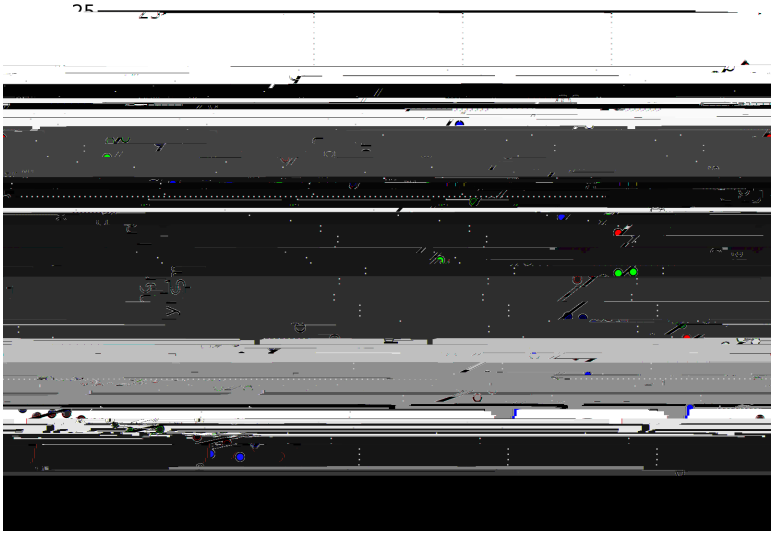


Figure 6.3: Estimating a best fit cubic for some random data; see Listing 6.3

```
x = arange(0.0, 2.0, 0.05)
```

```
im1 = imshow(Z1, cmap=cm.gray,
             interpolation='nearest', extent=extent)

# prevents the axes from clearing on next command
hold(True)

Z2 = func3(X, Y)
im2 =
```

6.4.4 Defining your own colormap

Per9y G9eenfield has provided a nice f9amework with `matplotlib.colors.LinearSegmentedColormap`

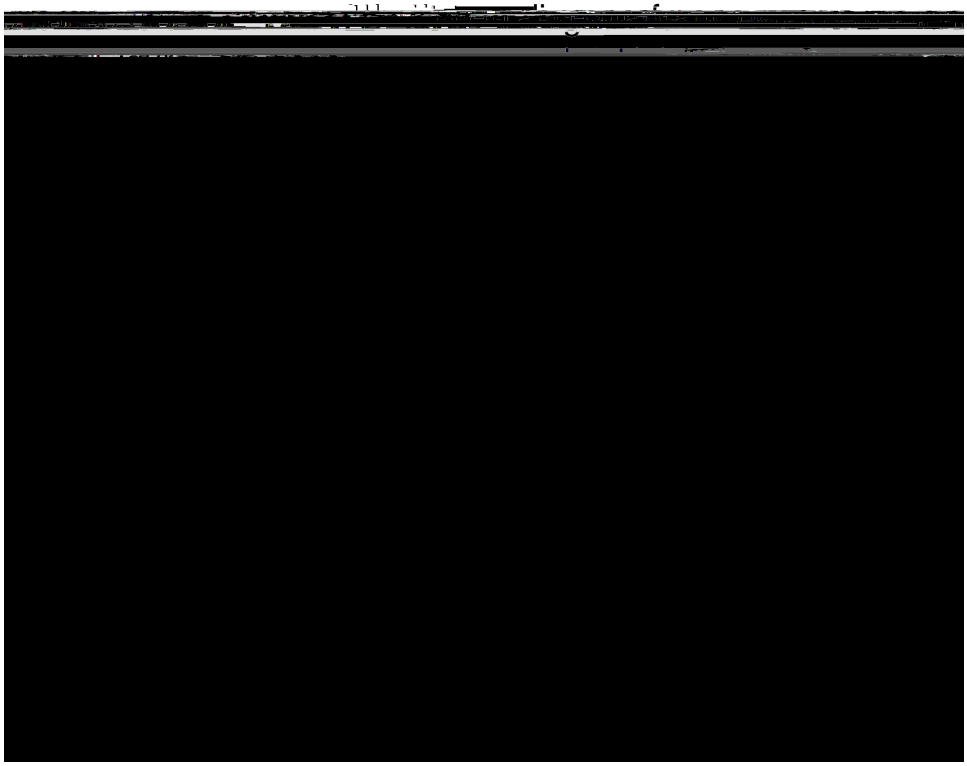


Figure 7.2: The inheritance diagram for The FigureCanvas hierarchy. The FigureCanvas is a backend

edei-11T89(a)-008GUIrwedSom(e)-008ofrtheave(c-27(k)28(ndi)1(s)-008we)-1inbpothrnvedra(n)1(d)400(a(n29(tigrain)1nd)]TJ

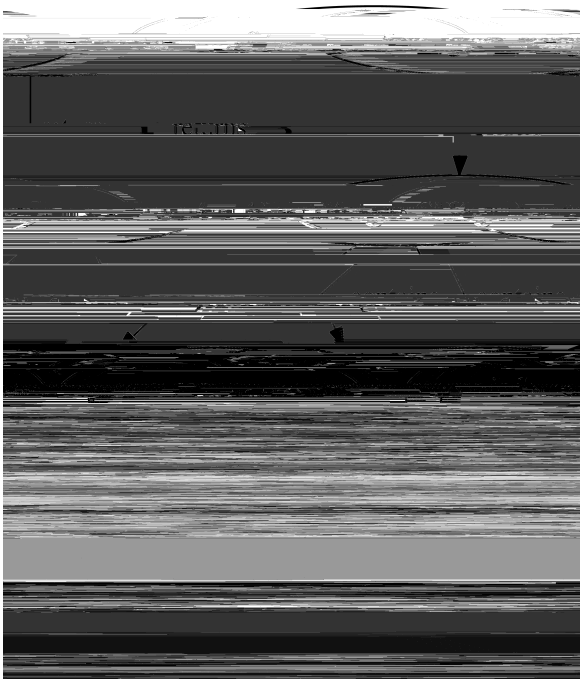


Figure 7.5: The pylab interface function new_

```
            manager.window.show()

        if gtk.main_level() == 0 and
```

Appendix A

A sample .matplotlibrc

```
####
```

```
lines.markeredgewidth   : 1.0
lines.markersize   : 6          # markersize,
```

```
font.stretch          : normal
font.size             : medium
font.serif            : New Century Schoolbook, Century Schoolbook L, Utopia, ITC
```

```
### SAVING FIGURES
# the
```

Appendix C

matplotlib source code license

All of the matplotlib src code is distributed under the Python Software Foundation (PSF) license, which permits commercial and noncommercial free use and redistribution as long as the conditions below are met. The VERSION string below is replaced by the current matplotlib version number with each release.

LICENSE AGREEMENT FOR MATPLOTLIB(of)-4nwith eacB(of)-4nnnn

THE POSSIBILITY THEREOF.

6. This License Agreement will automatically terminate upon a material breach of its terms and conditions.

7. Nothing in this License Agreement shall be deemed to create any relationship of agency, partnership, or joint venture between JDH and Licensee. This License Agreement does not grant permission to use JDH trademarks or trade name in a trademark sense to endorse or promote products or services of Licensee, or any third party.